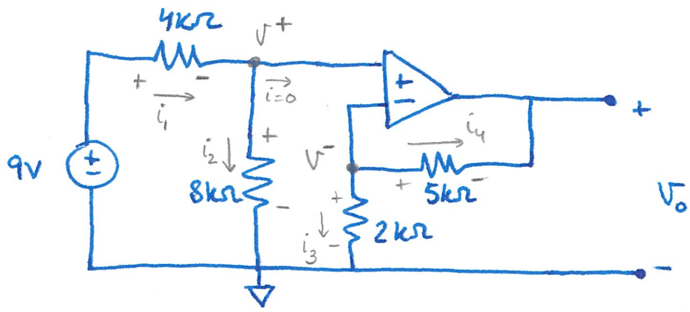
EXERCISE 5

Calculate V_o



1. KCL @ non-inverting input

$$i_1 - i_2 = 0$$

$$\frac{9 - V^+}{4\cancel{000}} - \frac{V^+}{8\cancel{000}} = 0 \Rightarrow \frac{9}{4} - \left(\frac{1}{4} + \frac{1}{8}\right)V^+ = 0 \; ; \; +\frac{3}{8}V^+ = +\frac{9}{4} \Rightarrow V^+ = \frac{9}{4} \times \frac{8}{3} = 6V$$

or use voltage divider: $V^+ = 9 \frac{8\cancel{K}}{8\cancel{K} + 4\cancel{K}} = \frac{72}{12} = 6V$

2. KCL @ inverting input

$$-i_3 - i_4 = 0$$

$$-\frac{V^-}{2\cancel{000}} - \frac{V^- - V_o}{5\cancel{000}} = 0 \Rightarrow -\left(\frac{1}{2} + \frac{1}{5}\right)V^- + \frac{V_o}{5} = 0 \; ; \; -\frac{7}{10}V^- + \frac{V_o}{5} = 0$$

Solve for V_o:

$$V_o = 5\,\frac{7}{10}V^- = \frac{7}{2}V^-$$

Since there is negative feedback: $V^+ = V^- = 6V$

Thus,

$$\boxed{V_o = \frac{7}{2}V^- = \frac{7}{2} \times 6 = 21V}$$